Waves at boundaries

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe reflection and transmission at a boundary.

SUCCESS CRITERIA - I can...

1. Identify incident, reflected and transmitted waves.
2. Distinguish fixed-end from free-end pulse reflection.
3. Track energy without assuming equal amplitudes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “boundary” and “reflection” to explain your model or result. Sentence starters are optional.

boundary: the place where a medium or constraint changes

reflection: a wave returning from a boundary

transmission: a wave continuing through or beyond a boundary

01 Warm-up: bring back an earlier idea

A reflected signal returns after 3 s.
Name the extra quantity needed to calculate distance and explain why this is a round trip.

Word help: signal = a disturbance that carries information

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

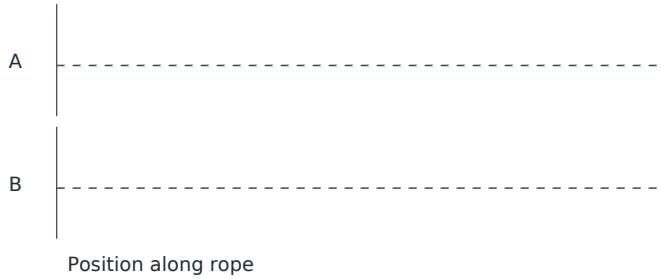
02 Worked example: explain the model

Reference: $f_{reflected} = f_{incident}$.
At a stationary boundary. A displacement pulse returns inverted from an ideal fixed end and upright from an ideal free end.

Worked reasoning: At a fixed end, an upward rope pulse reflects downward. At an ideal free end it reflects without inversion. These are different boundary conditions.

Your turn: Draw incident and reflected pulse snapshots for both ends; show propagation directions.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

An upward rope pulse reaches an ideal fixed end.
Sketch the reflected pulse and its travel direction.

Word help: ideal = using stated simplifying conditions

Sentence starter: The ... in my model represents ... . Its label shows ... .



Sketch and label your response.

Marzano check - circle: 1 2 3

04 Apply the physics

A pulse carries 10 J toward a boundary. The reflected pulse carries 4 J, and no energy is absorbed. Complete the energy-flow model and find the transmitted energy.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

Does a smaller transmitted amplitude prove frequency changed?
Explain what usually fixes the frequency at a stationary boundary.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 4 kg sample completely changes phase with latent heat 600 J/kg.
Find the energy magnitude required.
Label temperatures, samples and relevant energy transfers.

Word help: latent heat = energy per mass needed for a specified phase change

Sentence starter: I found ... [unit]. This value means ... in this situation.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

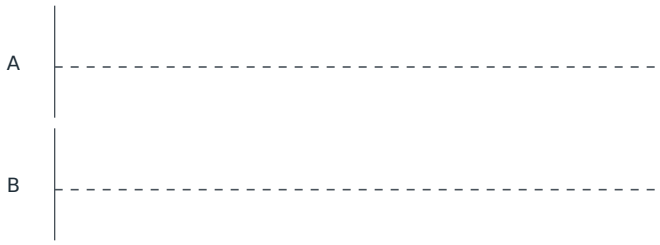
Sentence starters are optional. You may plan in any language.

07 Exit check

For an ideal free-end rope reflection, is an upward displacement pulse returned upright or inverted?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two equal upward pulses approach each other on an ideal rope.
Sketch overlap and later separation.
Explain whether they permanently stick.

Word help: ideal = using stated simplifying conditions

Sentence starter: My claim is … . The diagram/data show …, so … .



Sketch and label your response.

Marzano check - circle: 1 2 3